

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.18012	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.10186	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1487.586	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.073101	0.073101	271.58064	0.924659	14.96

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.18012.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.10186.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1487.586.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.068502.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073101.
- Held-out runtime ms: 271.58064.
- Robustness dispersion across held-out structure families: 0.009955.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.11394, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.098323, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.056154, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.052179, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.044909, errors=[]

Judge Note

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Phase-9 CVRP Solver-Evolution Suite Summary

- **Feasibility:**

All solvers, including the evolved solver and baselines, achieved perfect feasibility (100%) on held-out data.

- **Objective Gap (Held-out Feasible Gap):**

- Evolved solver: 0.0731
- Baseline Clarke-Wright Savings: 0.0738
- Baseline Nearest Neighbor: 0.2734
- Baseline Regret Insertion Local Search: 0.4664

The evolved solver has a slightly better gap than the best baseline (Clarke-Wright), but the difference is marginal.

- **Runtime (Held-out Runtime in ms):**

- Evolved solver: ~272 ms
- Clarke-Wright: ~77 ms (fastest)
- Nearest Neighbor: ~42 ms (fastest overall)
- Regret Insertion: ~1488 ms (slowest)

The evolved solver runs notably slower than the nearest neighbor and Clarke-Wright baselines but faster than regret insertion.

- **Robustness (Dispersion):**

- Evolved solver: 0.00996
- Clarke-Wright: 0.00779 (best)
- Nearest Neighbor: 0.00278 (lowest)
- Regret Insertion: 0.0869 (highest)

Robustness of the evolved solver is slightly worse than Clarke-Wright and nearest neighbor but much better than regret insertion.

- **Additional Notes:**

The evolved solver exhibits high code novelty (0.92) and additional complexity (≈ 15 nodes).

Conclusion

The phase-9 evolved solver attains feasibility equal to all baselines and a slightly improved objective gap compared to the best baseline (Clarke-Wright savings). However, this improvement in gap is marginal and comes at the cost of higher runtime and slightly reduced robustness. Therefore, the evolved solver **does not clearly and decisively outperform** the fixed baseline methods across held-out families but shows competitive performance with a trade-off in speed and complexity.